METAMIZOLE

Agranulocytosis risk

Banned in some countries

CHEMICAL NAME Sodium [(1,5-dimethyl-3-oxo-2-phenyl-2,3-dihydro-1H-pyrazol-4-yl)(methyl)amino]methanesulfonate

CLASS Non-opioid analgesic and antipyretic (pyrazolone derivative); also called dipyrone

TRADE NAMES Novalgin · Analgin · Baralgin M · Algozone

THERAPEUTIC USES / INDICATIONS

- Severe acute pain: post-operative, renal and biliary colic, cancer pain.
- High fever not responding to other measures.
- Spasmodic pain (has mild smooth-muscle relaxant effect).

PHARMACODYNAMICS

- Inhibits prostaglandin synthesis centrally and peripherally (COX inhibition).
- Strong analgesic and antipyretic; weak anti-inflammatory action.
- Mild spasmolytic effect on smooth muscle.

ADMINISTRATION

- **PO:** tablets or drops.
- **IV:** inject very slowly with the patient lying down (hypotension risk), or dilute in an infusion.
- **IM:** deep injection.
- Do not mix in the same syringe with other drugs.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, sweating.
- Red urine (harmless metabolite).
- Skin rash.

ADVERSE EFFECTS (SERIOUS)

- Agranulocytosis (rare, can be fatal): fever, sore throat, mouth ulcers.
- Hypotension and shock, especially with rapid IV injection.
- Anaphylaxis; SJS/TEN.
- Acute kidney injury (rare).

DOSAGES

- **Adult:** 500 mg–1 g PO/IV/IM up to 4 times daily; max 4 g/day.
- **Children:** 8–16 mg/kg per dose up to 4 times daily.
- Not recommended in infants <3 months or <5 kg.
- Use the lowest dose for the shortest time; reduce in renal or hepatic impairment.

CONTRAINDICATIONS & CAUTIONS

- Allergy to pyrazolones; history of agranulocytosis.
- Bone-marrow disorders; G6PD deficiency; acute porphyria.
- Third trimester of pregnancy; breastfeeding.
- **Caution:** hypotension, hypovolaemia, asthma with NSAID sensitivity.

KEY INTERACTIONS

- **Methotrexate:** ↑ bone-marrow toxicity.
- **Ciclosporin:** ↓ ciclosporin levels.
- **Chlorpromazine:** severe hypothermia.

PHARMACOKINETICS

- **Onset:** PO 30–60 min; IV within minutes. **Duration:** about 4 h.
- Prodrug: hydrolysed in the gut to the active metabolite 4-methylaminoantipyrine; metabolised in the liver.
- Excreted in urine (may colour urine red). **t½** of active metabolite: 2.5–3.5 h.

NURSING MANAGEMENT

- Ask about previous reactions to metamizole or other pyrazolones.
- Patient lying down for IV injection; inject slowly; monitor BP and pulse during and after.
- Watch for fever, sore throat or mouth ulcers → stop drug and check CBC.
- Long-term use: monitor CBC regularly.
- Assess pain and temperature response.

PATIENT EDUCATION

- Report fever, sore throat or mouth sores at once.
- Red-coloured urine is harmless.
- Do not use for longer than prescribed.
- Avoid during late pregnancy and breastfeeding.

KEY POINTS

- Strong analgesic/antipyretic; weak anti-inflammatory.
- Slow IV only; rapid injection → shock.
- Risk of agranulocytosis.
- Max 4 g/day in adults.

SOURCES CHECKED **[3]** EMA metamizole referral (2024)

Full references p. 432–457